

ZS-Q14

Canonical Kraus-Lift Selection by a Spin-1/2 Boundary

From Choi-Invisibility to Process-Level Falsification in Z-Spin Boundary Mediation

Kenny Kang June 2026 Theme: Quantum Channel Geometry / Quantum Mechanics / Quantum Hardware Interface | Paper Code: ZS-Q14 Version: v1.4 (supersedes v1.3)

Verification Summary: [zs_q14_verify_v1_4.py](#) — **33/33 PASS** across categories A–I (= the v1.3 suite's 30/30 plus a new Category I verifying the Boundary Spin-Lift Lemma). Central witness inherited verbatim from ZS-A7R Eq. (F.2) [34/34 PASS]. Zero new theoretical constants. **Status:**

TESTABLE-PROTOCOL / DERIVED-CONDITIONAL / VERIFIED-SIM / HARDWARE-PENDING

Primary Claim: A quantum channel alone does **not** determine a physical Kraus lift (Choi-invisibility No-Go + single-channel non-identifiability — both established and verified). If the Z-boundary physically carries the nontrivial (spinor) projective class of $SO(3)$ — i.e. $\dim Z = 2 = j = 1/2$ (ZS-M3 Thm 5.1; $Z = \partial X$, ZS-Q12) — then the half-angle $SU(2)$ connection is *forced* (Lemma 5.1, Boundary Spin-Lift), giving closure class $\chi_Z = -1$. The signed lift witness $\tilde{u}_Z(\theta)$ is a *process/dilation-level* observable, $\tilde{u}_Z = F(V, G_Z, \varphi_Z) \neq f(\Lambda)$, that kills ($\chi_Z = +1$) or survives ($\chi_Z = -1$) this selection principle.

Necessary-not-sufficient; **not** a new general theorem (the bundle/holonomy/No-Go/representation-theory results are prior art).

§0. Abstract

A completely positive trace-preserving (CPTP) channel is represented by Kraus operators only up to a unitary gauge freedom $K_a \mapsto \sum_b U_{ab} K_b$, $U \in U(r)$, and the Choi/Jamiołkowski matrix is the gauge-invariant that fixes the channel. We organize ZS-Q14 around a question meaningful to general quantum-information theory: **when does one lift of the Kraus gauge freedom cease to be a mere representation choice and become physical structure?**

Two facts bound the answer, and both are established or directly verified rather than claimed as new. (i) **Channel-Level Invisibility (No-Go, Theorem 1):** any function of the channel alone is invariant under closed loops in the Kraus-gauge fiber, so it cannot distinguish a 2π from a 4π closure; this is the gauge-invariance of the Choi matrix. (ii) **Single-Channel Non-Identifiability (Theorem 2):** the *same* channel yields closure class $\chi = -1$ under a half-angle lift $U_{1/2}(\theta) = e^{\{-i\theta\sigma_y/2\}}$ and $\chi = +1$ under a full-angle lift $U_1(\theta) = e^{\{-i\theta\sigma_y\}}$; the 4π is **injected by the chosen lift, not intrinsic** (verified to machine precision).

The contingent content is therefore not "one can see 4π " (textbook spinor/holonomy physics; Werner–Colella–Overhauser 1975, read as channel holonomy by Kult, Åberg & Sjöqvist 2007) but a **lift-selection criterion** (§4), and — new in v1.4 — a representation-theoretic ground for it. The **Boundary Spin-Lift Lemma** (Lemma 5.1) states that if the two-dimensional Z-boundary carries a faithful projective representation of $SO(3)$ in the nontrivial class of $H^2(SO(3), U(1)) \cong \mathbb{Z}_2$ (whose nontrivial 2π loop acts as $-I$), then the induced Kraus-frame connection is forced to be the half-angle generator $G = \sigma_y/2$ up to conjugation and orientation, whence $\chi_Z = -1$. This upgrades the criterion's connection condition (C3) from an independent assumption to a consequence of the premise that the Z-boundary is a physical spinor — exactly the Z-Spin postulate ($\dim Z = 2 = j = 1/2$; $Z = \partial X$). We are explicit that the lemma is elementary representation theory (the spin- $1/2$ representation is its canonical instance); its role is to tighten the paper's logic, not to claim new mathematics.

Z-Spin is thus a **candidate** lift-selector whose spinor-boundary postulate supplies the half-angle connection (§5). The signed lift witness $\tilde{u}_Z(\theta) = \text{Re Tr}[K_o(0)^\dagger K_o(\theta)]/\|K_o(0)\|F^2$ (ZS-A7R Eq. F.2) and the closure class $\chi_Z = \text{sgn}[\tilde{u}_Z(2\pi)/\tilde{u}_Z(0)]$ are the operational objects (§6). Crucially, $\tilde{u}_Z$ is **not a channel observable** — it is gauge-dependent — but a process/comb-level observable: with $\Lambda = \text{Tr}_Z \circ U_{XZ}$, one has $\tilde{u}_Z(\theta) \neq f(\Lambda)$ and $\tilde{u}_Z(\theta) = F(Y\{2:0\}, G_Z, R_Z) = F(V, G_Z, \phi_Z)$, where V is the Stinespring isometry, $G_Z = \sigma_y/2$ the selected connection, and ϕ_Z the environment phase reference exposed by the process tensor/comb $Y\{2:0\}$ but discarded by the channel marginal Λ (§7). The witness is read on the X-block of the OAQEC algebra $M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$, the Z-block being a scalar gauge subsystem (§8).

The experiment (§9) is a near-term, pre-registered, kill-or-survive test: $\chi_Z = +1$ refutes the spin- $1/2$ lift-selection, $\chi_Z = -1$ with the three load-bearing controls null is a *necessary* success. The same formalism applies to a second, independent physical channel — the hydrogenic hyperfine transition and its H / anti-H / muonium / positronium extensions (§10) — making ZS-Q14 a physical-channel-geometry program, not a single-circuit proposal.

Keywords: Kraus-gauge bundle, lift selection, boundary spin-lift, projective $SO(3)$ representation, channel holonomy, Choi/Jamiołkowski invariance, process tensor/comb, signed seam witness, closure class, spin-1/2 boundary, Z-Spin Cosmology, falsification test.

§0.1 Epistemic Status Legend

Status	Definition
PROVEN	Theorem inherited from prior Z-Spin papers or standard mathematics.
ELEMENTARY-KNOWN	True and provable here, already standard externally; stated only to locate the construction.
DERIVED	Follows from the Z-Spin action plus locked prior results, no new constants.

Status	Definition
DERIVED-CONDITIONAL	Derived under a stated physical postulate (here: $Z = \partial X / Z$ a physical spinor boundary).
CRITERION	A definition of when a contingent condition is taken to hold; not itself a derived theorem.
POSITION	A framing/localization statement, true and clarifying, not claimed as a new theorem.
TESTABLE-PROTOCOL	Concrete hardware experiment with pass/fail gates; not yet executed.
VERIFIED-SIM	Confirmed by classical simulation (zs_q14_verify_v1_4.py , inherited ZS-A7R suite).
HARDWARE-PENDING	Requires quantum-device execution.
CONTROL (load-bearing)	Negative experiment that <i>is</i> diagnostic.
CONTROL (non-diagnostic)	Negative experiment that, on analysis, does <i>not</i> separate the hypotheses.
NON-CLAIM	Explicitly not asserted.
OPEN	Well-posed unresolved issue.
RETRACTED	Previously circulated claim withdrawn.

§0.2 Position Statement

This paper's thesis is a **position**, not a new theorem: *a quantum channel does not determine a physical Kraus lift, but a physical boundary theory may*. The channel-holonomy field already operates under this fact — it introduces connections precisely because channels alone do not fix a holonomy (Uhlmann 1986; Kult, Åberg & Sjöqvist 2007). Z-Spin's contribution is a **specific candidate connection** (a spin- $\frac{1}{2}$ boundary), an experiment that kills or confirms it, and — in v1.4 — a representation-theoretic statement (Lemma 5.1) showing the half-angle connection is forced *given* that the boundary is a physical spinor.

This paper does **not** claim: a new general theorem about quantum channels; that the 4π closure is intrinsic to any channel (it is injected, §3); that $\tilde{u}_Z$ is a channel observable (it is gauge-dependent / process-level, §6–§7); that Lemma 5.1 is new mathematics (it is elementary representation theory, §5); any new quantum-computing algorithm, code, or primitive (the measurement is the standard Hadamard test); or that current hardware realizes the Z-sector. The one positive content is conditional: *if the Z-boundary is a physical spinor (the Z-Spin postulate), the half-angle lift is forced and $\tilde{u}_Z$ exhibits $\chi_Z = -1$; a 2π*

closure on hardware refutes the selection. The positive outcome is **necessary but not sufficient** for the full framework.

§1. The Lift-Selection Problem

In standard quantum-information theory the Kraus representation carries a $U(r)$ redundancy; the Choi/Jamiołkowski matrix is the gauge-invariant physical object. The channel-holonomy program (Kult, Åberg & Sjøqvist 2007), linked to Uhlmann's (1986) mixed-state holonomy via the Jamiołkowski isomorphism, studies the geometry of the Kraus $\rightarrow$ CPTP map and identifies a gauge-*invariant* holonomy as physical; the process-tensor framework (Pollock et al. 2018; Chiribella, D'Ariano & Perinotti 2009) carries more system–environment information than a single channel.

Against this backdrop the organizing question is:

When is a particular lift of the Kraus gauge freedom not a representation choice but physical structure?

The honest answer has three parts. The negative parts are established or verified (§3): a channel alone *cannot* select a lift (No-Go), and the *same* channel admits both a 2π and a 4π lift (non-identifiability). The constructive part (§4–§5) is a *criterion* sharpened by a representation-theoretic *lemma*: a physical theory whose boundary carries the nontrivial spinor class of $SO(3)$ forces the half-angle connection (Lemma 5.1) and, with an accessible dilation and bypass exclusion, selects the half-angle lift physically. Z-Spin is a candidate that supplies these via its spin- $\frac{1}{2}$ boundary. The experiment (§9–§10) tests that candidate.

§2. The Minimal Kraus Frame Bundle

Definition 1 (Minimal Kraus frame bundle). For a channel Λ of Kraus rank r , $P_\Lambda = \{ (K_0, \dots, K_{r-1}) : \sum_a K_a^\dagger K_a = I, \Lambda(\rho) = \sum_a K_a \rho K_a^\dagger \}$. Any two minimal frames are related by the gauge action $(K_a) \mapsto (\sum_b U_{ab} K_b)$, $U \in U(r)$; P_Λ is a $U(r)$ -torsor (a fiber). The assignment (frame) $\mapsto \Lambda$ is the bundle projection of Kult, Åberg & Sjøqvist (2007), with base invariant the Choi matrix $C_\Lambda = \sum_a |K_a\rangle\langle K_a|$.
[ELEMENTARY-KNOWN]

For $r = 2$ the fiber is $U(2)$; a *lift* is a path $\gamma(\theta) \in P_\Lambda$ generated by $\dot{K}(\theta) = -iG K(\theta)$ for a generator $G \in \mathfrak{u}(2)$ (a *connection*). Which connection, if any, is physically selected is the question of §1.

§3. Choi-Invisibility and Single-Channel Non-Identifiability

Theorem 1 (Channel-Level Invisibility No-Go) [ELEMENTARY-KNOWN]. Any function $f(\Lambda)$ of the channel alone — any process-tomography observable — is invariant under the entire $U(r)$ fiber, in particular under any closed loop $\theta \mapsto U(\theta)$. Hence no channel observable distinguishes a Kraus-fiber loop's 2π from 4π closure. *Proof.* Under $K_a \mapsto \Sigma_b U_{ab} K_b$, $C_\Lambda \mapsto \Sigma_{\{b,c\}} (U^\dagger U)_{cb} |K_b\rangle\langle K_c| = C_\Lambda$. (Verified to 3.3×10^{-16} ; suite Category C.)

Theorem 2 (Single-Channel Non-Identifiability) [DERIVED + VERIFIED-SIM]. Fix Λ . The closure class $\chi = \text{sgn}[\tilde{u}(2\pi)/\tilde{u}(0)]$ is **not** a function of Λ : under $G = \sigma_y/2$ it is -1 (period 4π), under $G = \sigma_y$ it is $+1$ (period 2π). The 4π is a property of the *chosen lift*. Equivalently the twist is a gauge no-op ($\Lambda_\theta = \Lambda$) and $\tilde{u}$ is gauge-dependent. *Proof.* Half-angle: $U(2\pi) = -I \Rightarrow \tilde{u}(2\pi) = -\tilde{u}(0)$. Full-angle: $U(2\pi) = +I \Rightarrow \tilde{u}(2\pi) = +\tilde{u}(0)$. (Verified: Category E — $\Lambda_\theta(\rho) = \Lambda(\rho)$ to 2.3×10^{-16} ; $r_{\text{half}} = -1$, $r_{\text{full}} = +1$; $\tilde{u}$ gauge-dependent.)

Theorem 2 forecloses any claim that the channel "hides" a 4π . This is what makes the constructive content of §4–§5 honest rather than numerological.

§4. The Physical Kraus-Lift Selection Criterion

We *define* when a lift is physically selected; we do not derive that any particular channel must be.

Criterion 4.1 (Physical lift selection) [CRITERION]. A connection G on the minimal frame bundle P_Λ of a rank-2 channel is *physically selected* when all four hold: **(C1) Minimal rank.** $r = \text{rank } C_\Lambda = \dim Z = 2$. **(C2) Accessible dilation.** In a Stinespring dilation $V : H_X \rightarrow H_X \otimes H_Z$, the environment H_Z carries an experimentally accessible phase reference (the Kraus index is a controllable boundary degree of freedom). **(C3) Fixed connection.** The half-angle generator $G = \sigma_y/2$ is fixed ($\dot{K}(\theta) = -i(\sigma_y/2) K(\theta)$). — *In v1.4 this is supplied by Lemma 5.1 below, not assumed.* **(C4) Bypass exclusion.** The direct $X \rightarrow Y$ path is absent or removed by load-bearing controls.

Proposition 4.1 (Closure-Class Connection-Invariance) [DERIVED + VERIFIED-SIM]. Given a fixed connection (C3), χ_Z is independent of the residual $U(2)$ gauge (the Kraus-frame anchor): $\chi_Z = -1$ for the half-angle generator and $+1$ for the full-angle generator, for *every* anchor. Hence χ_Z is a function of the connection, not of the gauge. *Proof.* Under $G = \sigma_y/2$, $U(2\pi) = -I$ gives $K_o(2\pi) = -K_o(0)$ for any anchor, so $\tilde{u}_Z(2\pi) = -\tilde{u}_Z(0)$; under $G = \sigma_y$, $\chi_Z = +1$. (Verified: Category F — NC2, 2000 anchors, all -1 = anchor-independence; Category E — connection-dependence.)

Honest boundary. Criterion 4.1 is not a theorem and Proposition 4.1 is modest: its substance — *a connection promotes a gauge-dependent holonomy to a connection-relative invariant* — is the channel-holonomy framework of Kult, Åberg & Sjöqvist (2007), building on Uhlmann (1986). What the criterion buys is clarity about which contingent inputs make the lift physical, hence exactly what the experiment falsifies.

§5. Z-Spin as a Candidate Spin-1/2 Lift Selector

Definition 2 (Z-Spin lift selector). $\Sigma_Z : (\Lambda, H_Z, J_Z, L_{XY} \equiv 0) \mapsto \gamma_Z(\theta) \in P_\Lambda$,
 $\gamma_Z(\theta) = e^{\{-i\theta\sigma_y/2\}} \cdot (K_0, K_1)$: not an arbitrary gauge path but the path selected by the Z-sector conditions.

The connection condition (C3) is now a representation-theoretic consequence, not a free assumption:

Lemma 5.1 (Boundary Spin-Lift Lemma). [*Representation-theory content:*

ELEMENTARY-KNOWN. Use here: DERIVED-CONDITIONAL.] Let $H_Z \cong \mathbb{C}^2$ carry a faithful projective unitary representation of $SO(3)$ in the **nontrivial class** of $H^2(SO(3), U(1)) \cong \mathbb{Z}_2$ — equivalently, a representation under which the generator of a nontrivial 2π loop in $\pi_1(SO(3)) \cong \mathbb{Z}_2$ acts as $-I$. Then (i) the representation is unitarily equivalent to the spin- $1/2$ ($j = 1/2$) representation of $SU(2) = \text{Spin}(3)$; (ii) the induced connection on the rank-2 Kraus frame of a dilation with environment H_Z is generated by a half-angle generator $G = n \cdot \sigma/2$ for some unit vector n , i.e. $G \cong \sigma_y/2$ up to conjugation (axis) and orientation (sign); (iii) consequently $\chi_Z = -1$. Conversely, a representation in the **trivial class** ($2\pi \mapsto +I$) gives the full-angle (vector) connection and $\chi_Z = +1$.

Proof. $H^2(SO(3), U(1)) \cong \mathbb{Z}_2$ (Bargmann 1954); the nontrivial class consists of the genuine (linear) representations of the universal double cover $SU(2)$. The unique 2-dimensional irreducible representation of $SU(2)$ is the spin- $1/2$ representation, with generators $J_k = \sigma_k/2$; a rotation by θ about axis n is $e^{\{-i\theta n \cdot \sigma/2\}}$, and the lift of a 2π rotation is the path $I \rightarrow -I$ in $SU(2)$, acting as $D^{\{1/2\}}(-I) = -I$ (ZS-M3 Lemma 10.1). Faithfulness as a projective $SO(3)$ representation follows from $PU(2) \cong SO(3)$. Since every $n \cdot \sigma$ has eigenvalues ± 1 , a unitary conjugation sends $n \cdot \sigma \mapsto \sigma_y$, hence $G = n \cdot \sigma/2 \cong \sigma_y/2$ up to conjugation and orientation. The closure class follows from Theorem 2. The converse (trivial class) factors through $SO(3)$ with $2\pi \mapsto +I$, giving the full-angle generator and $\chi_Z = +1$. (Verified: suite Category I — $2\pi \mapsto -I$ for 200 random axes to 1.2×10^{-16} ; $n \cdot \sigma/2 \sim \sigma_y/2$ by explicit unitary to 8.9×10^{-16} ; spinor/vector dichotomy.)

Effect on the criterion (logical economy). With Lemma 5.1, (C3) is no longer an independent assumption: it is forced by the premise that H_Z physically carries the nontrivial spinor class. That premise is exactly the Z-Spin postulate — $\dim Z = 2 =$ the *unique* $j = 1/2$ spinor subspace (ZS-M3 Theorem 5.1, PROVEN), anchored as the boundary $Z = \partial X$ (ZS-Q12, DERIVED-CONDITIONAL). Hence the spinor-boundary postulate now implies (C1) [2-dim] and (C3) [half-angle connection] *jointly*,

leaving only **(C2)** [accessible dilation] and **(C4)** [bypass exclusion] as operational/experimental conditions. The remaining Z-Spin inputs:

- **(C2)** The Z-sector is the physical Planck-scale boundary $H_Z = \partial(H_X)$; the dilation environment is the Z-boundary with the seam phase as reference (ZS-Q12).
- **(C4)** $L_{XY} \equiv 0$ (ZS-F5, PROVEN) is bypass exclusion at the level of the action; on hardware it is enforced by the load-bearing controls (§9).

Honest boundary. Lemma 5.1 is textbook representation theory — the spin- $\frac{1}{2}$ representation is its canonical example, and $H^2(SO(3), U(1)) \cong \mathbb{Z}_2$ is standard (Bargmann 1954; Hall 2015). It introduces no new mathematics; its contribution is to the paper's logical economy, deriving the half-angle connection from the already-postulated spinor nature of the Z-boundary rather than positing it separately. The contingency (and the falsifiability) is unchanged: *if* the boundary is a physical spinor, $\chi_Z = -1$; a hardware $\chi_Z = +1$ refutes that premise.

§6. The Signed Lift Witness and the Closure Class

Definition 3 (Signed lift witness). $\tilde{u}_Z(\theta) = \text{Re Tr}[K_o(0)^\dagger K_o(\theta)] / \|K_o(0)\|_{F^2}$, $K_o(\theta) = [e^{\{-i\theta\sigma_y/2\}}]_{ou} K_u(0) = \cos(\theta/2)K_o(0) - \sin(\theta/2)K_i(0)$. (= ZS-A7R Eq. F.2.)

Definition 4 (Closure class). $\chi_Z = \text{sgn}[\tilde{u}_Z(2\pi)/\tilde{u}_Z(0)] \in \{-1, +1\}$; $\chi_Z = -1 \Leftrightarrow$ spinorial (half-angle) lift, $+1 \Leftrightarrow$ vector (full-angle) lift.

Properties (ZS-A7R Theorem 3.2-bis, PROVEN; suite Category B): $\tilde{u}_Z(0) = +1$; $\tilde{u}_Z(\theta) = \cos(\theta/2)$ for seam-orthogonal Kraus; $\tilde{u}_Z(\theta+2\pi) = -\tilde{u}_Z(\theta)$; $\tilde{u}_Z(\theta+4\pi) = +\tilde{u}_Z(\theta)$; uniqueness as the Kraus-linear discriminator (given the half-angle structure). $\tilde{u}_Z$ is *linear* in $K_o(\theta)$ and inherits the $j = 1/2$ sign flip.

Why a bilinear/probability witness fails (RETRACTED). A probability contrast $W = P(0|2\pi) - P(0|4\pi)$ is bilinear and θ -constant by Theorem 1 (verified to 3.3×10^{-16} , Category D); the θ -trivial "double-J echo" channel is likewise retracted (Category D). The witness is the linear $\tilde{u}_Z$; the discriminant is the sign χ_Z — never an absolute amplitude, never equal to A.

§7. Process-Level Identifiability and Experimental Access [POSITION — formalized]

Theorems 1–2 say $\tilde{u}_Z$ is invisible at the channel level and gauge-dependent. The correct formal statement of *what it is* uses the process tensor / quantum comb (Chiribella, D'Ariano & Perinotti 2009; Pollock et al. 2018). Let $Y_{\{2:0\}}$ denote the two-time process comb of the controlled dilation — the higher-order map taking the inserted control operation on the environment H_Z (here the twist $e^{\{-i\theta G_Z\}}$) to the output system state. The channel is the zero-control marginal:

$\Lambda = \text{Tr}_Z \circ U_{XZ}$, $U_{XZ}(\cdot) = V(\cdot)V^\dagger$, $V : H_X \rightarrow H_X \otimes H_Z$ the Stinespring isometry.

The signed lift witness is **not** a function of Λ :

$$\tilde{u}_Z(\theta) \neq f(\Lambda), \quad \tilde{u}_Z(\theta) = F(Y_{\{2:0\}}, G_Z, R_Z) = F(V, G_Z, \phi_Z),$$

where V is the Stinespring isometry (the dilation), $G_Z = \sigma_y/2$ is the selected connection (Lemma 5.1), and ϕ_Z is the environment phase reference R_Z — the control degree of freedom that the comb $Y_{\{2:0\}}$ exposes but the channel marginal Λ discards. Concretely, $F(V, G_Z, \phi_Z) = \text{Re Tr}[K_0(0)^\dagger K_0(\theta)]/\|K_0(0)\|_{F^2}$ with $K_a(\theta)$ the Kraus operators of V twisted by $e^{i\theta G_Z}$, the reference ϕ_Z fixing the anchor $K_0(0)$.

The " $\neq f(\Lambda)$ " is exactly the gauge-dependence verified in Theorem 2 / suite Category E: two dilations V, V' of the *same* Λ give different $\tilde{u}_Z$. The content: **$\tilde{u}_Z$ is a process-level observable of the comb $Y_{\{2:0\}}$, recoverable only with control of the dilation (the environment phase reference ϕ_Z)** — the multi-time/dilation information the comb carries beyond Λ . Operationally this is a Hadamard test on the Kraus index (ZS-QH §6.1; Nielsen–Chuang) with the untwisted anchor $K_0(0)$ as reference beam — the quantum-information analogue of the Werner–Colella–Overhauser (1975) reference beam that Kult, Åberg & Sjøqvist (2007) read as an early channel-holonomy realization. ZS-Q14 is therefore a *reduced comb/dilation tomography* of a physically-selected lift, not channel tomography. (This is a localization; the underlying fact that dilation phases require dilation access is Uhlmann 1986 — no new theorem is claimed.)

§8. OAQEC Localization: Why the Witness is Read on X, not Z

ZS-Q11 (Theorem Q11.A, PROVEN) gives the single-cell logical algebra $A_{\text{ZS}} \cong M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$, code $\dim 9 = 1 + 3 + 5$ (suite Category G reproduces this). The Z-logical block is the scalar $\mathbb{C}$ — a *gauge subsystem* with trivial logical algebra — so $\tilde{u}_Z$ is **not** a Z-logical observable; it is read on the X-block $M_3(\mathbb{C})$ under the Z-frame error set $E_Z = \{I_X \oplus V \oplus I_Y\}$, against which X-logical information is correctable (Theorem Q11.B). The witness probes J_Z-graded transport *through* the Z-gauge subsystem, detected on X — consistent with its being a process/dilation observable (§7). Following Dauphinais, Kribs & Vasmer (2024) (OAQEC stabilizer formalism characterizing correctable errors), and per ZS-Q11's own non-claim (NC-XIV.4), this is recognition of external mathematics, not a new Z-Spin claim.

§9. Hardware Protocol and Load-Bearing Controls (Track B0)

Encoding. Embed $\{X: x_0, x_1, x_2\}$, $\{Z: z_0, z_1\}$, $\{Y: y_0, \dots, y_5\}$ into a 16-dim four-qubit space (5 leakage states); $p_{\text{leak}} = 1 - \text{Tr}(P_{\text{code}} \rho P_{\text{code}}) < 1\%$ ($< 5\%$ feasibility-only).

Circuit. (1) Prepare $|\psi_X\rangle$, ancilla $|+\rangle$. (2) Inject $X \rightarrow Z$ (the 2-channel Kraus index = dilation environment). (3) Controlled half-angle twist $e^{-i\theta\sigma_Y/2}$ on the Kraus index, $\theta \in \{0, 2\pi, 4\pi\}$ + fine sweep. (4) Hadamard-test readout of $\tilde{u}_Z(\theta)$. (5) Form $\chi_Z = \text{sgn}[\tilde{u}_Z(2\pi)/\tilde{u}_Z(0)]$. (6) Run all controls. Cost $\sim 8\%$ shots over ZS-A4 KS-2.

Negative controls (three load-bearing, two non-diagnostic; suite Category F).

- **NC1 — Bypass** (removes the 2-channel Z transit $\rightarrow$ removes the half-angle): $\chi_Z = +1$ (null). Tests (C4).
- **NC4 — Phase-only** (full- vs half-angle generator): $\chi_Z = +1$ (null). Tests (C3) / Lemma 5.1.
- **NC5 — Z-off** (no twist): $\chi_Z = +1$ (null). Tests (C2); a residual X-response to Y-stimulation independently violates $L_{XY} \equiv 0$.
- **NC2 — Random involution / NC3 — Label shuffle:** $\chi_Z = -1$ (do **not** null), because they perturb J_Z / labels but not the half-angle connection (Proposition 4.1). **Non-diagnostic** — recorded as diagnostics, never as gates.

Decision rule. Pre-register $N_s \geq 12$ input states, $\geq 10^4$ shots, $p < 0.01$ after correction, bootstrap CIs, TOST on the load-bearing controls only. Pass: $\chi_Z = -1$ at $> 5\sigma$; NC1/NC4/NC5 inside ROPE ($\chi \approx +1$); p_{leak} below threshold; reproduced on ≥ 2 backends or one backend + classical noise-injection audit. Native qudit/qutrit (Track A0) gives $p_{\text{leak}} \approx 0$; a material-stack realization (Track C0; ZS-QH FMDs) is future work.

§10. The Hydrogenic Hyperfine Channel as a Second Physical Target

The lift-selection formalism is not specific to engineered hardware; the same Σ_Z applies to a second, independent physical channel, which is why ZS-Q14 is a *physical-channel-geometry program* rather than a single-circuit proposal.

ZS-Q13 treats the hydrogen hyperfine transition as a Z-Spin-mediated rank-2 channel. There the *unsigned* Choi seam witness vanishes in the ideal limit ($u_{\text{seam}} \rightarrow 0$) — a channel-level (bilinear) quantity, blind to the lift by Theorem 1 — while the *signed* lift witness $\tilde{u}_Z$ must carry the Kraus-level 4π closure ($\chi_Z = -1$) if the spin- $\frac{1}{2}$ boundary selects the half-angle lift (Lemma 5.1). This is the same Choi-invisible / process-visible split as §3–§7, realized in atomic physics.

The discriminating power grows across the hydrogenic family. A program over **H / anti-H / muonium / positronium** separates proton-structure, CPT, and leptonic systematics, because the Σ_Z prediction ($\chi_Z = -1$, a parameter-free sign) is common while the conventional backgrounds differ by system. Two independent physical instances of the *same* closure-class prediction — an engineered four-qubit channel and the hydrogenic hyperfine channel — constitute a far stronger test of the lift-selection principle than either alone. (Status inherited from ZS-Q13; not re-derived here.)

§11. Falsification Gates

Gate	Condition	Consequence
F-Q14.1	$\chi_Z = +1$ (2π / vector) on the Z-mediated channel	spin-½ lift-selection refuted — primary kill
F-Q14.2	NC1 bypass gives $\chi_Z = -1$	effect not tied to Z transit ((C4) fails)
F-Q14.4	NC4 phase-only gives $\chi_Z = -1$	ordinary phase, not the half-angle connection ((C3)/Lemma 5.1 fails)
F-Q14.5	NC5 Z-off gives a signed signal	twist not located in Z ((C2) fails)
F-Q14.6	leakage exceeds threshold	invalid protocol
F-Q14.7	backend-to-backend reproduction fails	hardware artifact
F-Q14.9	Z-off gives X-response to Y-stimulation	direct X–Y path (violates $L_{XY} \equiv 0$)
F-Q14.10	no sim/hardware agreement after calibration	model invalid
F-Q14.11	hydrogenic $\chi_Z = +1$ (ZS-Q13 target)	selection principle fails in the second physical instance

A single failure among F-Q14.1, F-Q14.2, F-Q14.4, F-Q14.5, F-Q14.6 blocks any claim that the lift is physically selected.

§12. Non-Claims and Prior-Art Boundary

This paper does **not** claim: (1) a new general theorem about quantum channels — the bundle (Kult, Åberg & Sjöqvist 2007), mixed-state holonomy (Uhlmann 1986), Choi/Jamiołkowski invariance (Choi 1975; Jamiołkowski 1972), the No-Go (Theorem 1), the process-comb localization (Chiribella et al. 2009; Pollock et al. 2018), and the representation theory of Lemma 5.1 (Bargmann 1954; Hall 2015) are all prior art; (2) that quantum channels generically hide a 4π topology (Theorem 2); (3) that $\tilde{u}_Z$ is a channel observable (§6–§7); (4) any new quantum-computing algorithm, code, or primitive; (5) that Criterion 4.1 is a derived theorem, or that Lemma 5.1 is new mathematics (it is textbook representation theory used for logical economy); (6) that the selection is unconditional (DERIVED-CONDITIONAL on Z being a physical spinor boundary, $Z = \partial X$); (7) that NC2/NC3 are diagnostic; (8) that a positive result proves the full Z-Spin framework (necessary, not sufficient); (9) that current hardware realizes the Z-sector or the full Topological Defect Controller. The single contingent positive content is the *candidate* spin- $\frac{1}{2}$ lift-selection (§5) and its falsifiable prediction $\chi_Z = -1$.

§13. Conclusion

A quantum channel cannot, by itself, select a Kraus lift (Theorem 1, known), and the same channel admits both a 2π and a 4π lift (Theorem 2, verified): the 4π is injected, not intrinsic. What can select a lift is a *physical connection* on the Kraus bundle — the established channel-holonomy fact — together with an accessible dilation. v1.4 grounds the connection condition representation-theoretically: if the Z -boundary carries the nontrivial spinor class of $SO(3)$, the half-angle generator $\sigma_y/2$ is forced (Lemma 5.1), so the spinor-boundary postulate alone implies both the minimal rank (C1) and the half-angle connection (C3). Z-Spin is then a candidate selector whose spin- $\frac{1}{2}$ boundary makes $\chi_Z = -1$ a falsifiable, parameter-free, sign-valued prediction; the signed lift witness that measures it is a process/comb-level observable $\tilde{u}_Z = F(V, G_Z, \phi_Z) \neq f(\Lambda)$ (§7), read on the X -block of the OAQEC algebra (§8), testable on near-term hardware (§9) and in the hydrogenic hyperfine channel (§10). The paper is a proposal — a *selection principle* with a representation-theoretic spine — rather than a defense: a 2π result kills it, a 4π result with controls null is a necessary success, and the framing claims no mathematics that is not either Z-Spin's own postulate or established prior art.

Acknowledgements & Code Availability

Consolidated under Kenny Kang's editorial direction. The verification suite [zs_q14_verify_v1_4.py](#) (33/33 PASS, categories A–I) and the inherited [ZS_A7_v1_0_verification.py](#) (34/34, ZS-A7R) reproduce all quantitative claims of v1.4. No new theoretical constants; (A, Q, dim Z) = (35/437, 11, 2) LOCKED.

Appendix A. Circuit Notation

Encoded basis $\{|0\rangle, \dots, |10\rangle\} = X_3 \oplus Z_2 \oplus Y_6$; $J_Z: |z_0\rangle \leftrightarrow |z_1\rangle$ (Z -internal, distinct from antipodal J : $|j\rangle \mapsto |Q-1-j\rangle$; $\langle J, J_Z \rangle \cong D_4$). Connection generator $G = \sigma_y/2$ (half-angle, forced by Lemma 5.1 under the spinor-boundary premise), lift $U(\theta) = e^{\{-i\theta\sigma_y/2\}}$. Witness by the Hadamard-test estimator of $\tilde{u}_Z(\theta) = \text{Re Tr}[K_0(\theta)^\dagger K_0(\theta)]/\|K_0(\theta)\|_F^2$, $K_0(\theta) = \cos(\theta/2)K_0(0) - \sin(\theta/2)K_1(0)$; closure class $\chi_Z = \text{sgn}[\tilde{u}_Z(2\pi)/\tilde{u}_Z(0)]$. **Do not** use the bilinear probability witness (θ -constant; Thm 1, Category D) or the double-J echo channel (θ -trivial; Category D).

Appendix B. Pre-Registration Template

Fix before running: (1) backend; (2) date/calibration snapshot; (3) encoding map; (4) J_Z and the connection generator $G = \sigma_y/2$; (5) the three load-bearing controls (NC1, NC4, NC5) and ROPE; (6) angle set $\{0, 2\pi, 4\pi\} + \text{fine sweep}$; (7) shot count; (8) input-state set ($N_s \geq 12$); (9) leakage threshold; (10) test (bootstrap + TOST on load-bearing controls only); (11) gates of §11. NC2/NC3 recorded as diagnostics, not gates. Post-hoc changes are exploratory.

Appendix C. Verification (mapping to v1.4)

[zs_q14_verify_v1_4.py](#), seed 20260601, numpy only, **33/33 PASS** (A 4/4 | B 5/5 | C 3/3 | D 2/2 | E 4/4 | F 5/5 | G 4/4 | H 3/3 | **I 3/3**). Mapping: **A** well-posedness; **B** Definition 3 properties; **C** Theorem 1; **D** §6 retractions; **E** Theorem 2 (gauge no-op; χ connection-dependent; $\tilde{u}_Z$ gauge-dependent — the " $\neq f(\Lambda)$ " of §7); **F** Criterion 4.1 / Proposition 4.1 (NC1/NC4/NC5 null; NC2 2000 anchors all -1 ; NC3 -1); **G** §8 OAQEC ($(Z, X, Y) = (1, 3, 5)$, Z -block scalar; reproduces ZS-Q11 Thm Q11.A); **H** noise; **I** — **Lemma 5.1 (Boundary Spin-Lift)**: the nontrivial 2π loop acts as $-I$ for 200 random axes (1.2×10^{-16}); every half-angle generator $n \cdot \sigma/2$ is unitarily conjugate to $\sigma_y/2$ (8.9×10^{-16}); spinor $\rightarrow -I$ ($\chi = -1$) vs vector $\rightarrow +I$ ($\chi = +1$) dichotomy. Key residuals $\leq 4.4 \times 10^{-16}$.

Appendix D. Deep-Exploration Record (v1.4)

- **Adopted (both feedback items, judged sound)**: (1) the **Boundary Spin-Lift Lemma** (Lemma 5.1), which upgrades condition (C3) from an independent assumption to a representation-theoretic consequence of the spinor-boundary premise, so that the postulate now implies (C1) + (C3) jointly; verified by the new suite Category I. (2) the **formalization of §7** via the process $\text{comb } Y_{\{2:0\}}$ and the explicit $\tilde{u}_Z(\theta) = F(V, G_Z, \phi_Z) \neq f(\Lambda)$, with V the Stinespring isometry, $G_Z = \sigma_y/2$ the connection, ϕ_Z the environment phase reference.
- **Honest guardrails**: Lemma 5.1 is elementary representation theory (the spin- $1/2$ representation; $H^2(\text{SO}(3), U(1)) \cong \mathbb{Z}_2$); it adds no new mathematics and is tagged accordingly — its role is the paper's logical economy. The §7 formalization is a POSITION/localization (the dilation-access

fact is Uhlmann 1986), not a new theorem. Verification grows $30 \rightarrow 33$ with the genuinely new Category I; no claim is verified beyond what the simulations show.

References

Internal (Z-Spin Cosmology):

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Version History

v1.4, June 2026 (this version): Adds **Lemma 5.1 (Boundary Spin-Lift Lemma)** — if H_Z carries a faithful projective $SO(3)$ representation in the nontrivial class of $H^2(SO(3), U(1)) \cong \mathbb{Z}_2$ ($2\pi \mapsto -I$), the induced Kraus-frame connection is the half-angle generator $\sigma_y/2$ up to conjugation/orientation — which upgrades criterion condition (C3) from an independent assumption to a representation-theoretic consequence of the spinor-boundary postulate (the postulate now implies (C1) + (C3) jointly).

Formalizes §7 with the process $\text{comb } Y_{\{2:0\}}$ and the explicit $\tilde{u}_Z(\theta) = F(Y_{\{2:0\}}, G_Z, R_Z) = F(V, G_Z, \phi_Z) \neq f(\Lambda)$ (V Stinespring isometry, $G_Z = \sigma_y/2$ connection, ϕ_Z environment phase reference). Verification suite grows $30 \rightarrow 33/33$ with a new Category I ([zs_q14_verify_v1_4.py](#)). Honest guardrails retained: Lemma 5.1 is elementary representation theory (no new mathematics; logical-economy role), the §7 formalization is a localization (Uhlmann 1986), and no new general theorem or quantum-computing contribution is claimed. All v1.3 content retained (lift-selection reframing; Definitions 1–4; Theorems 1–2; Criterion 4.1 / Proposition 4.1; candidate-selector framing; hydrogenic second target), as are all v1.1 corrections (linear $\tilde{u}_Z$; bilinear-witness and double-J-echo retractions; OAQEC $M_3 \oplus \mathbb{C} \oplus M_5$ with the witness on X ; corrected controls; "Z-Spin-mediated" terminology). Positive outcome **necessary but not sufficient**. ($A, Q, \dim Z = (35/437, 11, 2)$) LOCKED.

v1.3, June 2026 (superseded): Recentered the paper on the lift-selection question; added Definitions 1–4, Theorem 1 (No-Go), Theorem 2 (non-identifiability), Criterion 4.1 and Proposition 4.1, §5 (Z-Spin as candidate selector), §7 (process-level localization), and §10 (hydrogenic second target); recast the feedback's two "theorems" as a criterion + conditional proposition and a position.

v1.1, June 2026 (superseded): Added the Kraus-gauge-bundle formalism and the Choi-invisibility proposition with external attribution; established by simulation the gauge no-op, injected 4π , and gauge-dependence; corrected the OAQEC dimension (M_5) and control structure; replaced the bilinear witness with the linear $\tilde{u}_Z$ and retracted the double-J echo.

v1.0, June 2026 (superseded): Initial consolidation into a near-term falsification test.